

Research Interests

As an AI researcher focused on accelerating scientific progress, I believe that scalable and impactful scientific systems equipped with autonomy are essential. I build these systems through a unified framework that integrates reasoning agents—grounded in scientific laws and equipped with truthful tools—with large-scale generative models acting as scientific world models. Crucially, operating this unified framework at a real-world-meaningful scale requires a seamless architecture driven by deep HPC and system-level CUDA optimizations. This synergy of agentic reasoning and high-performance optimization enables robust closed-loop exploration, ultimately paving the way for long-term visions like autonomous laboratories.

Topics AI for Science; Foundation Models; High-Performance Computing; Scientific Agents; Large Language Models; Multimodal Reasoning.

Education & Research

Mar. 2023 – **Ph.D. Candidate**, *Korea University*, Seoul, Korea

Present Major: Artificial Intelligence

Advisor: Sungwoong Kim, Sungbin Lim

Mar. 2021 – **M.S.**, *UNIST*, Ulsan, Korea

Feb. 2023 Major: Artificial Intelligence

Advisor: Sungbin Lim

May 2019 – **Military Service**, *Korean Navy*, Korea

Jan. 2021

Dec. 2016 – **Research Intern**, *UNIST*, Ulsan, Korea

Jun. 2018 Advisor: Cheolmin Ghim

Subject: Markov Chain Monte Carlo Simulation of Complex Networks

Dec. 2015 – **Research Intern**, *UNIST*, Ulsan, Korea

Jun. 2016 Advisor: Sangkyu Kwak

Subject: Molecular Dynamics Simulation of Desalination using Carbon Nanotube

Mar. 2014 – **B.S.**, *UNIST*, Ulsan, Korea

Feb. 2019 Major: Chemical Engineering

Industrial Experience

Apr. 2025 – **Industry-Academia Project**, *LG AI Research*, Seoul, Korea

Mar. 2026 Materials Intelligence Lab. — Chemical Reasoning Agent: Multi-turn Molecular Optimization and Generation with Tool Augmentation

Dec. 2023 – **Industry-Academia Project (Project Lead)**, *LG AI Research*, Seoul, Korea

Dec. 2024 Materials Intelligence Lab. — Multimodal Foundation LLM for Molecular Chemistry

Sep. 2023 – **Research Intern (Full-time)**, *LG AI Research*, Seoul, Korea

Sep. 2024 Materials Intelligence Lab. — Chemical LLM Agent: Data Generation, Training, and Evaluation using MLOps with In-house Deployment

Apr. 2023 – **Industry-Academia Project (Project Lead)**, *LG AI Research*, Seoul, Korea

Apr. 2024 Data Intelligence Lab. — LLM Causal Reasoning for Causal Discovery

Jun. 2022 – **Research Intern (Full-time)**, *LG AI Research*, Seoul, Korea

Nov. 2022 Materials Informatics Lab. — Bayesian Inference Applied to Wet Lab Chemistry Data in a Data-Scarce Regime

Publications

(* denotes equal contribution or co-corresponding)

○ **Mol-LLM: Multimodal Generalist Molecular LLM with Improved Graph Utilization.**

Chanhui Lee, Hanbum Ko, Yuheon Song, Yongjun Jeong, Rodrigo Hormazabal, Sehui Han, Kyunghoon Bae, Sungbin Lim*, Sungwoong Kim*.

Under review at Nature Communications.

○ **RetroReasoner: A Reasoning LLM for Strategic Retrosynthesis Prediction.**

Hanbum Ko, Chanhui Lee, Ye Rin Kim, Rodrigo Hormazabal, Sehui Han, Sungbin Lim, Sungwoong Kim.

Under review at EMNLP 2026.

○ **MolDesignBench: Evaluating LLM-based Agent for Scenario-grounded Molecular Design.**

Yongjun Jeong, Hanbum Ko, Ye Rin Kim, Chanhui Lee, Rodrigo Hormazabal, Jaewan Lee, Sehui Han, Sungbin Lim, Sungwoong Kim.

Under review at COLM 2026.

○ **Machine Learning Hamiltonians are Accurate Energy-Force Predictors.**

Seongsu Kim, Chanhui Lee, Yoonho Kim, Seongjun Yun, Honghui Kim, Nayoung Kim, Changyoung Park, Sehui Han, Sungbin Lim*, Sungsoo Ahn*.

ICML 2026.

○ **Score-informed Neural Operator for Enhancing Ordering-based Causal Discovery.**

Jiyeon Kang*, Songseong Kim*, Chanhui Lee, Doyeong Hwang, Joanie Hayoun Chung, Yunkyoung Ko, Sumin Lee, Sungwoong Kim, Sungbin Lim.

NeurIPS 2025.

○ **ReactionReasoner: Towards Reasoning LLM for Chemical Reaction Prediction.**

Hanbum Ko, Chanhui Lee, Ye Rin Kim, Rodrigo Hormazabal, Sehui Han, Sungbin Lim, Sungwoong Kim.

NeurIPS Workshop on AI for Science 2025.

○ **Mol-LLM: Multimodal Generalist Molecular LLM with Improved Graph Utilization.**

Chanhui Lee, Hanbum Ko, Yuheon Song, Yongjun Jeong, Rodrigo Hormazabal, Sehui Han, Kyunghoon Bae, Sungbin Lim*, Sungwoong Kim*.

NeurIPS Workshop on AI for Science 2025.

○ **On Incorporating Prior Knowledge Extracted from Pre-trained Language Models into Causal Discovery.**

Chanhui Lee*, Juhyeon Kim*, Yongjun Jeong, Yoonseok Yeom, Juhyun Lyu, Jung-Hee Kim, Sangmin Lee, Sangjun Han, Hyeokjun Choe, Soyeon Park, Woohyung Lim, Kyunghoon Bae, Sungbin Lim, Sanghack Lee.

NeurIPS Workshop on Causality and Large Models 2024.

○ **Scalable Multi-Task Transfer Learning for Molecular Property Prediction.**

Chanhui Lee, Dae-Woong Jeong, Sung Moon Ko, Sumin Lee, Hyunseung Kim, Soorin Yim, Sehui Han, Sungwoong Kim, Sungbin Lim.

ICML Workshop on AI4Science 2024.

○ **Task Addition in Multi-Task Learning by Geometrical Alignment.**

Soorin Yim, Dae-Woong Jeong, Sung Moon Ko, Sumin Lee, Hyunseung Kim, Chanhui Lee, Sehui Han.

ICML Workshop on AI4Science 2024.

○ **Multitask Extension of Geometrically Aligned Transfer Encoder.**

Sung Moon Ko, Sumin Lee, Dae-Woong Jeong, Hyunseung Kim, Chanhui Lee, Soorin Yim, Sehui Han.

ICML Workshop on AI4Science 2024.

- **Design of high-performance entangling logic in silicon quantum dot systems with Bayesian optimization.**
Jihoon Kang, Taehyun Yoon, Chanhui Lee, Sungbin Lim, Hoon Ryu.
Scientific Reports 14, 10080 (2024).

———— **Academic Activities**

Conference NeurIPS (2023, 2024, 2025 Top Reviewer); ICML (2023, 2024, 2025); ICLR (2023, 2024,
Reviewer 2025); ACL (2025); AAAI (2024); TMLR (2025).